Music Analyser & Procedural Dancer Animation: Technical Blueprint

1. Executive Summary & Review of Claude's Feedback

Claude's critique is an incisive, pragmatic engineering assessment. In an interactive browser-based visualizer running at 60 FPS, attempting to execute large multimodal machine learning pipelines (such as diffusion transformers conditioned on Jukebox or MERT embeddings) introduces prohibitive computational latency, high VRAM overhead, and licensing restrictions. Generative dance models typically require seconds of offline computation to synthesize short movement chunks, whereas a real-time visualizer requires low-latency, frame-by-frame reactivity.

The optimal approach is to extract the verified biomechanical, cognitive, and somatic principles from dance science and translate them into a layered, procedural animation engine powered by standard Web Audio API nodes.

2. Fact Verification of Flagged Items

A. Frequency Bands (50–100 Hz and 6,400–12,800 Hz)

- **Status: Fully Verified.**
- In the empirical motion capture research conducted by Birgitta Burger et al. (*"Influences of Rhythm- and Timbre-Related Musical Features on Characteristics of Music-Induced Movement"*, Frontiers in Psychology), audio was decomposed into 10 octave-scale sub-bands via the MATLAB MIRtoolbox.
- **Sub-band 2 (50–100 Hz):** Captures kick drum fundamentals and bass synthesizer lines. It was shown to induce vertical Center of Mass (CoM) excursion, hip sway, and sagittal knee flexion via saccular-vestibular excitation.
- **Sub-band 9 (6,400–12,800 Hz):** Captures cymbals, open hi-hats, and high-frequency percussive transients. It correlates with rapid, low-mass kinematic adjustments in distal extremities (hands, wrists, and head turns).

B. PopDanceSet Dataset Volume

- **Status: Claude is Correct; Figure Corrected.**
- The POPDG paper (*"POPDG: Popular 3D Dance Generation with PopDanceSet"*, CVPR) records PopDanceSet duration at **12,819 seconds (~3.56 hours)** across 19 genres and 132 subjects. The 9.0-hour figure in the earlier literature survey conflated PopDanceSet with an older commercial capture dataset.

C. Motion Capture Dataset Licensing Restrictions

- **Status: Claude is Correct.**
- Academic benchmarks such as AIST++ and FineDance are strictly licensed under non-commercial research terms. Pre-baked animations extracted from them cannot be legally redistributed in commercial web software or public broadcast streaming without a dedicated license. Any production MoCap clips must be commissioned, self-captured, or obtained from commercially cleared repositories (such as Mixamo or CC0 public domain conversions).

3. Step-by-Step Procedural Implementation Blueprint

Step 1: Decouple Body Architecture into Two Independent Layers

Dance biomechanics reveals a natural functional bifurcation: the lower body acts as a metric stabilization engine, while the upper body acts as a melodic and gestural interpreter. Tying arms and legs together into monolithic full-body keyframes causes movement to feel rigid and combinatorial variety to collapse.
Compute joint transformations additively:

$$T_{final}(j,t) = T_{lower}(j,t) \oplus \left(w_j \cdot T_{upper}(j,t)\right)$$

- **Lower Unit (Hips, Knees, Ankles):** $w_j = 0.0$. Governed strictly by metric pulse, bounce, and support.
- **Core / Spine / Ribcage:** $w_j \approx 0.5 - 0.7$. Blends weight transfers with chest undulations.
- **Upper Unit (Shoulders, Wrists, Head):** $w_j = 1.0$. Driven by melodic gestures and high-frequency transients.

Combinatorial Benefit: Authoring 4 procedural lower-body footwork loops and 6 upper-body gestural routines immediately produces $4 \times 6 = 24$ distinct dance variations without needing 24 hand-crafted animations.

Step 2: Automated Motion Test Harness (Eliminating Eyeball Guesses)

Implement programmatic unit tests to grade animation quality numerically across three criteria:

1. **Kinematic Beat Turnaround:** A choreographic beat occurs at local kinetic velocity minima (stopping points and direction reversals), not peak velocity. Compute the mean skeletal joint speed $v(t)$:
   $$v(t) = \frac{1}{\vert{}J\vert{}} \sum_{j \in J} \frac{\Vert{}\mathbf{p}_j(t) - \mathbf{p}_j(t - \Delta t)\Vert{}_2}{\Delta t}$$
   Flag frames where $v(t) < v(t - \Delta t)$ and $v(t) < v(t + \Delta t)$. If $\vert{}t - t_{\text{beat}}\vert{} > 50\text{ ms}$, mark the move as rhythmically desynchronized.
   Physical Foot Slide (Skate) Penalty: During contact phases, horizontal foot speed must clamp to zero ($< 0.05 m/s$). Evaluate the Physical Foot Contact (PFC) error:

$$SlideError(t) = \| a_{CoM}(t) \|_2 \cdot \| v_{L_foot}(t) \|_2 \cdot \| v_{R_foot}(t) \|_2$$

2. Non-zero values indicate phantom acceleration without physical ground traction.
Dynamic Center of Mass (CoM) Stability: Calculate projected 2D ground CoM. Verify that CoM stays bounded within the Base of Support (BoS):

$$\min(x_{L_foot}, x_{R_foot}) - \delta \leq x_{CoM} \leq \max(x_{L_foot}, x_{R_foot}) + \delta$$

3. (where δ is a dynamic margin allowing brief ballistic leanings during steps).

Step 3: Surgical Web Audio API Feature Extraction

Acoustic Feature	Web Audio Implementation	Target Motion Parameter
Sub-Bass (50–100 Hz)	BiquadFilter (bandpass, center 75 Hz, Q = 1.5)	Knee bounce amplitude, vertical CoM excursion, hip sway.
High Frequencies (6.4–12.8 kHz)	BiquadFilter (bandpass, center 9,600 Hz, Q = 1.5)	Distal transients: wrist flicks, head tilt impulses.
Percussive Snap (Slope)	Derivative of the RMS envelope follower: $\frac{dRMS}{dt}$	Laban "Sudden" effort; reduces damping factor to create crisp stops.
Pulse Clarity Metric	$1.0 - \min\left(1.0, \frac{StdDev(\Delta t_{beats})}{Mean(\Delta t_{beats})}\right)$	Clarity > 0.8 drives periodic stepping; Clarity < 0.4 triggers arm-searching behavior.

Step 4: Procedural Mathematical Formulation for the House "Jack"

The house jack is a continuous sagittal wave rippling upward through knees, pelvis, chest, and head. To render it readable on a 2D front-facing figure lacking depth, project sagittal curvature into phase-staggered vertical offsets and perspective scaling:
Let the normalized beat phase cycle be $\phi \in [0, 2\pi]$ across each musical beat:

$$\begin{aligned} y_{\text{pelvis}}(\phi) &= -A_{\text{pelvis}} \cdot \sin(\phi) \\ y_{\text{chest}}(\phi) &= -A_{\text{chest}} \cdot \sin\left(\phi - \frac{\pi}{4}\right) \\ y_{\text{head}}(\phi) &= -A_{\text{head}} \cdot \sin\left(\phi - \frac{\pi}{2}\right) \end{aligned}$$

Front-Perspective Visual Cues:

1. **Thoracic Extension:** Dynamically increase distance $(y_{neck} - y_{pelvis})$ during chest extension to simulate upward torso opening.
2. **Perspective Expansion:** Widen shoulder width $(x_{R_shoulder} - x_{L_shoulder})$ by 3% to 5% at peak forward push to simulate forward optical perspective.
3. **Clavicle Articulation:** Lift shoulder markers vertically as the chest reaches its crest, then drop them sharply on beat arrival.

Step 5: Engineering Execution Strategy

- **Phase 1 (Immediate):** Run a 100% procedural, layered animation engine driven by Web Audio API filter nodes inside a standard requestAnimationFrame render loop.
- **Phase 2 (Automated Testing):** Integrate the mathematical checks (Kinematic Turnaround, PFC Foot Slide, and BoS Balance) into headless test suites to validate procedural choreography without subjective visual inspection.
- **Phase 3 (Optional MoCap Primitives):** If pre-recorded motion is desired later, capture or license short 2-bar loops under clear commercial licenses (e.g., house two-step, shuffle, salsa rock). Apply phase-vocoder time warping to match loop playback rates to detected BPM, while retaining real-time procedural upper-body responsiveness.